The Basics

Lists are represented with the bracket [] syntax. You can easily create them, add to them, and work with them just as in Ruby. Enter the following code in an IEx session:

```
$ iex
iex> list = [1, 2, 3]
[1, 2, 3]

iex> list ++ [4, 5]
[1, 2, 3, 4, 5]

iex> list -- [2, 3]
[1]

iex> [0 | list]
[0, 1, 2, 3]

iex> length(list)
3
```

++ concatenates the second list onto the end of the first. This is equivalent to list + list in Ruby.

-- removes the first occurrence of the right list from the left. This is similar to list - list in Ruby but differs from it because Ruby removes all occurrences and Elixir only removes a single occurrence.

[element | list] adds the given element to the beginning of a list. This is useful for performance reasons, which we'll cover shortly.

The List[1] module provided by Elixir has many utility functions in it. One function that seems to be missing from the module is a function that gets the length of a list. Elixir provides length/1 in the Kernel module, so you can invoke length(list) directly.

Tuples

Tuples appear to be similar to lists but are used completely differently. Tuples are fixed-size containers that are efficiently stored in memory. Enter the following code in an IEx session to see them in action:

```
$ iex
iex> tuple = {:ok, 1}
{:ok, 1}

iex> tuple_size(tuple)
2
```

1. https://hexdocs.pm/elixir/List.html